

15.20 abc

2026-06-01

15.20 This exercise uses data from the STAR experiment introduced to illustrate fixed and random effects for grouped data. In the STAR experiment, children were randomly assigned within schools into three types of classes: small classes with 13-17 students, regular-sized classes with 22-25 students, and regular-sized classes with a full-time teacher aide to assist the teacher. Student scores on achievement tests were recorded as well as some information about the students, teachers, and schools. Data for the kindergarten classes are contained in the data file star.

```
load("/Users/liaochenghao/Desktop/Financial Econometrics/Homework/HW8/Data/star.rdata")
df <- star
names(df)
```

```
## [1] "id"          "schid"       "tchid"       "tchexper"    "absent"
## [6] "readscore"   "mathscore"   "totalscore"   "boy"         "white_asian"
## [11] "black"       "tchwhite"    "tchmasters"   "freelunch"   "schurban"
## [16] "schrural"    "small"       "regular"      "aide"
```

- a. Estimate a regression equation (with no fixed or random effects) where *READSCORE* is related to *SMALL*, *AIDE*, *TCHEXPER*, *BOY*, *WHITE_ASIAN*, and *FREELUNCH*. Discuss the results. Do students perform better in reading when they are in small classes? Does a teacher's aide improve scores? Do the students of more experienced teachers score higher on reading tests? Does the student's sex or race make a difference?

```
model_a <- lm(readscore ~ small + aide + tchexper + boy + white_asian + freelunch, data = df)
summary(model_a)
```

```
##
## Call:
## lm(formula = readscore ~ small + aide + tchexper + boy + white_asian +
##     freelunch, data = df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -107.220  -20.214   -3.935   14.339  185.956
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  437.76425    1.34622  325.180 < 2e-16 ***
## small         5.82282     0.98933   5.886 4.19e-09 ***
## aide          0.81784     0.95299   0.858  0.391
## tchexper      0.49247     0.06956   7.080 1.61e-12 ***
## boy          -6.15642     0.79613  -7.733 1.23e-14 ***
## white_asian   3.90581     0.95361   4.096 4.26e-05 ***
## freelunch    -14.77134     0.89025 -16.592 < 2e-16 ***
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 30.19 on 5759 degrees of freedom
## (20 observations deleted due to missingness)
## Multiple R-squared:  0.09685,    Adjusted R-squared:  0.09591
## F-statistic: 102.9 on 6 and 5759 DF,  p-value: < 2.2e-16
```

I estimate a pooled OLS regression of READSCORE on SMALL, AIDE, TCHEXPER, BOY, WHITE_ASIAN, and FREELUNCH, without fixed effects or random effects. The estimated regression equation is

$$READ\hat{SCORE} = 437.7643 + 5.8228SMALL + 0.8178AIDE + 0.4925TCHEXPER - 6.1564BOY + 3.9058WHITE_ASIAN - 14.7713FREELUNCH \quad (1)$$

The coefficient on *SMALL* is 5.8228 and is statistically significant at the 1% level. This suggests that, holding other factors constant, students in small classes score about 5.82 points higher on reading tests than students in regular-sized classes. Therefore, students appear to perform better in reading when they are assigned to small classes.

The coefficient on *AIDE* is 0.8178, but it is not statistically significant. Thus, there is no strong statistical evidence that having a full-time teacher aide improves reading scores relative to a regular-sized class without an aide.

The coefficient on *TCHEXPER* is 0.4925 and is statistically significant. This means that one additional year of teacher experience is associated with an increase of about 0.49 points in reading scores. Therefore, students taught by more experienced teachers tend to score higher on reading tests.

The coefficient on *BOY* is -6.1564 and is statistically significant, indicating that boys score about 6.16 points lower than girls, holding other variables constant.

The coefficient on *WHITE_ASIAN* is 3.9058 and is statistically significant, suggesting that White and Asian students score about 3.91 points higher than other students.

Finally, the coefficient on *FREELUNCH* is -14.7713 and is highly statistically significant. This indicates that students receiving free lunch score about 14.77 points lower than students who do not receive free lunch.

Overall, small class size, teacher experience, gender, race, and free-lunch status are statistically significant predictors of reading scores, while the teacher aide variable is not statistically significant.

-
- b. Reestimate the model in part (a) with school fixed effects. Compare the results with those in part (a). Have any of your conclusions changed? [*Hint: specify SCHID as the cross-section identifier and ID as the “time” identifier.*]

```
# id = individual/student identifier, used as the "time" index here
pdata <- pdata.frame(df, index = c("schid", "id"))

model_b <- plm(readscore ~ small + aide + tchexper + boy + white_asian + freelunch, data = pdata, model = "fixed", index = "id")

## Warning in index.pindex(x = anindex, which = which): an index variable not
## being the individual index is called 'id'. Likely, any results are distorted.
## Warning in index.pindex(x = anindex, which = which): an index variable not
## being the individual index is called 'id'. Likely, any results are distorted.
## Warning in index.pindex(x = anindex, which = which): an index variable not
## being the individual index is called 'id'. Likely, any results are distorted.
```

```
summary(model_b)
```

```
## Warning in index.pindex(x = anindex, which = which): an index variable not
## being the individual index is called 'id'. Likely, any results are distorted.
## Warning in index.pindex(x = anindex, which = which): an index variable not
## being the individual index is called 'id'. Likely, any results are distorted.
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## being the individual index is called 'id'. Likely, any results are distorted.
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## being the individual index is called 'id'. Likely, any results are distorted.
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## being the individual index is called 'id'. Likely, any results are distorted.
## Warning in index.pindex(x = anindex, which = which): an index variable not
## being the individual index is called 'id'. Likely, any results are distorted.
## Warning in index.pindex(x = anindex, which = which): an index variable not
## being the individual index is called 'id'. Likely, any results are distorted.
```

```
## Oneway (individual) effect Within Model
```

```
##
```

```
## Call:
```

```
## plm(formula = readscore ~ small + aide + tchexper + boy + white_asian +
##       freelunch, data = pdata, effect = "individual", model = "within")
```

```
##
```

```
## Unbalanced Panel: n = 79, T = 34-137, N = 5766
```

```
##
```

```
## Residuals:
```

```
## Warning in index.pindex(x = anindex, which = which): an index variable not
## being the individual index is called 'id'. Likely, any results are distorted.
```

```
##      Min. 1st Qu.  Median 3rd Qu.    Max.
```

```
## -102.64  -16.78   -2.85   12.76  198.42
```

```
##
```

```
## Coefficients:
```

```
##              Estimate Std. Error  t-value Pr(>|t|)
## small          6.490231   0.912962   7.1090 1.313e-12 ***
## aide            0.996087   0.881693   1.1297  0.2586
## tchexper        0.285567   0.070845   4.0309 5.629e-05 ***
## boy            -5.455941   0.727589  -7.4987 7.440e-14 ***
## white_asian     8.028019   1.535656   5.2277 1.777e-07 ***
## freelunch     -14.593572   0.880006 -16.5835 < 2.2e-16 ***
```

```
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
## Warning in index.pindex(x = anindex, which = which): an index variable not
## being the individual index is called 'id'. Likely, any results are distorted.
```

```
## Total Sum of Squares:    4628000
```

```
## Warning in index.pindex(x = anindex, which = which): an index variable not
## being the individual index is called 'id'. Likely, any results are distorted.
```

```
## Residual Sum of Squares: 4268900
## R-Squared:      0.077592
## Adj. R-Squared: 0.063954
## F-statistic: 79.6471 on 6 and 5681 DF, p-value: < 2.22e-16
```

I reestimate the model using school fixed effects, where SCHID is used as the cross-section identifier and ID is used as the time identifier. The school fixed effects model controls for all unobserved school-level characteristics that are constant within a school.

The estimated fixed effects results are:

$$READ\hat{SCORE} = 6.4902SMALL + 0.9961AIDE + 0.2856TCHEXPER - 5.4559BOY + 8.0280WHITE_ASIAN - 14.5936FREE\hat{LUNCH} \quad (2)$$

Compared with the pooled OLS results in part (a), the coefficient on *SMALL* increases from 5.8228 to 6.4902 and remains statistically significant. Thus, the conclusion that students in small classes perform better in reading does not change.

The coefficient on *AIDE* increases slightly from 0.8178 to 0.9961, but it remains statistically insignificant. Therefore, there is still no evidence that a teacher aide significantly improves reading scores.

The coefficient on *TCHEXPER* decreases from 0.4925 to 0.2856 but remains statistically significant. This suggests that students taught by more experienced teachers still tend to have higher reading scores, although the estimated effect is smaller after controlling for school fixed effects.

The coefficient on *BOY* remains negative and statistically significant, indicating that boys score lower than girls in reading. The coefficient on *WHITE_ASIAN* remains positive and statistically significant, and it becomes larger after including school fixed effects. The coefficient on *FREE\hat{LUNCH}* remains negative and highly significant, with a magnitude very similar to the pooled OLS estimate.

Overall, the main conclusions from part (a) do not change after including school fixed effects. Small classes improve reading scores, teacher aides do not have a statistically significant effect, teacher experience has a positive effect, boys score lower than girls, White/Asian students score higher, and students receiving free lunch score lower.

-
- c. Test for the significance of the school fixed effects. Under what conditions would we expect the inclusion of significant fixed effects to have little influence on the coefficient estimates of the remaining variables?

```
# Pooled OLS model
model_pool_lm <- lm(readscore ~ small + aide + tchexper + boy + white_asian + freelunch, data = df)

# Model with school fixed effects
model_fe_lm <- lm(readscore ~ small + aide + tchexper + boy + white_asian + freelunch + factor(schid), data = df)

# F-test for school fixed effects
anova(model_pool_lm, model_fe_lm)
```

```
## Analysis of Variance Table
##
## Model 1: readscore ~ small + aide + tchexper + boy + white_asian + freelunch
## Model 2: readscore ~ small + aide + tchexper + boy + white_asian + freelunch +
##      factor(schid)
##   Res.Df    RSS Df Sum of Sq    F    Pr(>F)
## 1     5759 5247584
```

```
## 2    5681 4268894 78    978690 16.698 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

To test the significance of the school fixed effects, I compare the pooled OLS model with the school fixed effects model using an F -test.

The null hypothesis is that the school fixed effects are not jointly significant, meaning that all school effects are equal. The alternative hypothesis is that the school fixed effects are jointly significant.

The ANOVA test gives an F -statistic of 16.698 with a p-value less than $2.2e-16$. Since the p-value is much smaller than 0.05, I reject the null hypothesis. Therefore, the school fixed effects are jointly statistically significant.

This result suggests that there are important differences across schools that affect students' reading scores. These differences may reflect school quality, school resources, neighborhood characteristics, or other school-level factors.

However, significant fixed effects do not necessarily imply that the coefficient estimates of the remaining variables will change substantially. The coefficient estimates will be little affected if the explanatory variables are approximately uncorrelated with the school fixed effects.

In the STAR experiment, students were randomly assigned within schools to small classes, regular classes, or regular classes with a teacher aide. Therefore, the treatment variables *SMALL* and *AIDE* should not be strongly correlated with unobserved school-level characteristics. As a result, even though the school fixed effects are statistically significant, including them has relatively little effect on the coefficient estimates for these variables.